

Arthur Cayley

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607. A Memoir on Prepotentials (concluded)

[Continued from earlier pages of the same paper. Articles 131 (concl.)–162.]

[p. 406] V' is not infinite at any point outside the cylinder; and it vanishes at infinity, except indeed when only the coordinate e is infinite, and its form at infinity is not

$$= M(a^2 + \cdots + c^2 + e^2)^{-\frac{1}{2}(s-2)}.$$

V'' is not infinite for any point within the cylinder; and at the surface we have $V' = V''$.

We have

$$\rho = -\frac{\Gamma(\frac{1}{2}s - \frac{1}{2})}{4(\Gamma\frac{1}{2})^{s+1}} \left(\frac{dW'}{da^2} + \frac{dW''}{da'^2} \right),$$

where

$$\frac{dW'}{da^2} = \frac{-(s-2)\frac{1}{2}(a^2 + \cdots + z^2)M}{(a^2 + \cdots + c^2)^{\frac{1}{2}s}}, \quad \frac{dW''}{da'^2} = \frac{-(s-2)M}{f^{s-2}} \text{ at the surface; } \frac{dW''}{da'^2} = 0,$$

and therefore

$$\rho = \frac{(s-2)\Gamma(\frac{1}{2}s - \frac{1}{2})f^{s-1}M}{4(\Gamma\frac{1}{2})^{s+1}f^{s-1}} \quad (\text{viz. } \rho \text{ is constant});$$

or, what is the same thing, writing $M = \frac{4(\Gamma\frac{1}{2})^{s+1}f^{s-1}\rho}{(s-2)\Gamma(\frac{1}{2}s - \frac{1}{2})}$, whence $\rho = Bf$, and writing also $a^2 + \cdots + c^2 = \varpi^2$, we have

$$V' = \frac{4(\Gamma\frac{1}{2})^{s+1}f^{s-1}\rho}{(s-2)\Gamma(\frac{1}{2}s - \frac{1}{2})} \cdot \frac{1}{\varpi^{s-2}} \quad \text{for an exterior point } \varpi > f,$$

$$V'' = \frac{4(\Gamma\frac{1}{2})^{s+1}f^{s-1}\rho}{(s-2)\Gamma(\frac{1}{2}s - \frac{1}{2})} \cdot \frac{1}{f^{s-2}} \quad \text{for an interior point } \varpi < f.$$

132. This is right; but we can without difficulty bring it to coincide with the result obtained for the $(s+1)$ -dimensional sphere with only $s-1$ in place of s ; we may in fact, by a single integration, pass from the cylinder $x^2 + \cdots + z^2 = f^2$ to the s -dimensional sphere or circle $x^2 + \cdots + z^2 = f^2$, which is the base of this cylinder. Writing first $dS = d\Sigma dw$, where $d\Sigma$ refers to the s variables $(x, \dots, z)$ and the sphere $x^2 + \cdots + z^2 = f^2$; or using now dS in this sense, then in place of the original dS we have $d\Sigma dw$: and w being ∞ , the limits of w being $\pm\infty$, then in place of $e - w$ we may write simply w . This being so, and putting for shortness $(a-x)^2 + \cdots + (c-z)^2 = A^2$, the integral is

$$V = \iint \frac{\delta f dS dw}{(A^2 + w^2)^{\frac{1}{2}(s-2)}},$$

and we have without difficulty

$$\int_{-\infty}^{+\infty} \frac{dw}{(A^2 + w^2)^{\frac{1}{2}(s-2)}} = \frac{1}{A^{s-3}} \cdot \frac{\sqrt{\pi} \Gamma\frac{1}{2}(s-2)}{\Gamma\frac{1}{2}(s-1)}.$$

[p. 407] To prove it, write $w = A \tan \theta$, then the integral is in the first place converted into

$$\frac{2}{A^{s-3}} \int_0^{\frac{\pi}{2}} \cos^{s-4} \theta d\theta,$$

which, putting $\cos \theta = \sqrt{x}$ and therefore $\sin \theta = \sqrt{1-x}$, becomes

$$= \frac{1}{A^{s-3}} \int_0^1 x^{\frac{1}{2}s-\frac{5}{2}} (1-x)^{-\frac{1}{2}} dx,$$

which has the value in question.

Hence, replacing A by its value, we have

$$V = \frac{\sqrt{\pi} \Gamma_{\frac{1}{2}}(s-2)}{\Gamma_{\frac{1}{2}}(s-1)} \int \frac{\delta f dS}{\{(a-x)^2 + \dots + (c-z)^2\}^{\frac{1}{2}(s-3)}} = \frac{4\pi^{\frac{1}{2}s} \Gamma(\frac{1}{2}) f^{s-1} \rho}{(s-2) \Gamma_{\frac{1}{2}}(s-2)} \cdot \frac{1}{(a^2 + \dots + c^2)^{\frac{1}{2}(s-3)}} \quad \text{or} \quad \frac{1}{f^{s-3}};$$

that is,

$$V = \frac{4(\Gamma_{\frac{1}{2}})^s f^{s-1} \rho}{(s-3) \Gamma_{\frac{1}{2}}(s-3)} \cdot \frac{1}{(a^2 + \dots + c^2)^{\frac{1}{2}(s-3)}} \quad \text{or} \quad \frac{1}{f^{s-3}};$$

viz. this is the formula for the sphere with $s-1$ instead of s .

Annex VII. Example of Theorem D, Art. Nos. 133 and 134.

133. The example relates to the $(s+1)$ -dimensional sphere $x^2 + \dots + z^2 + w^2 = f^2$.

Instead of at once assuming for V a form satisfying the proper conditions as to continuity, we assume a form with indeterminate coefficients, and make it satisfy the conditions in question. Write

$$V = \frac{M}{(a^2 + \dots + c^2 + e^2)^{\frac{1}{2}(s-1)}} \quad \text{for } a^2 + \dots + c^2 + e^2 > f^2;$$

$$V = A(a^2 + \dots + c^2 + e^2) + B \quad \text{for } a^2 + \dots + c^2 + e^2 < f^2.$$

In order that the two values may be equal at the surface, we must have

$$\frac{M}{f^{s-1}} = Af^2 + B;$$

in order that the derived functions $\frac{dV}{da}$, &c. may be equal, we must have

$$\frac{-(s-1)aM}{f^{s+1}} = 2Aa, \quad \&c.,$$

viz. these are all satisfied if only $\frac{-(s-1)M}{f^{s+1}} = 2A$.

We have thus the values of A and B ; or the exterior potential being as above

$$V = \frac{M}{(a^2 + \dots + c^2 + e^2)^{\frac{1}{2}(s-1)}},$$

[p. 408] the value of the interior potential must be

$$V = -\frac{M}{f^{s+1}} \left\{ \left(\frac{1}{2}s + \frac{1}{2} \right) - \left(\frac{1}{2}s - \frac{1}{2} \right) \frac{a^2 + \dots + c^2 + e^2}{f^2} \right\}.$$

The corresponding values of W are of course

$$\frac{M}{(a^2 + \dots + c^2 + e^2 + w^2)^{\frac{1}{2}(s-1)}} \quad \text{and} \quad \frac{M}{f^{s+1}} \left\{ \left(\frac{1}{2}s + \frac{1}{2} \right) - \left(\frac{1}{2}s - \frac{1}{2} \right) \frac{a^2 + \dots + c^2 + e^2 + w^2}{f^2} \right\};$$

and we thence find

$$\begin{aligned} \rho &= 0, & \text{if } a^2 + \dots + c^2 + w^2 > f^2, \\ \rho &= -\frac{\Gamma(\frac{1}{2}s - \frac{1}{2})}{4(\Gamma(\frac{1}{2})^{s+1})} \left\{ -4(\frac{1}{2}s - \frac{1}{2})(\frac{1}{2}s + \frac{1}{2}) \frac{M}{f^{s+3}} \right\} = \frac{\Gamma(\frac{1}{2}s + \frac{1}{2})}{(\Gamma(\frac{1}{2})^{s+1})} \cdot \frac{M}{f^{s+3}}, \\ & & \text{if } a^2 + \dots + c^2 + w^2 < f^2. \end{aligned}$$

Assuming for M the value $\frac{(\Gamma(\frac{1}{2})^{s+1})}{\Gamma(\frac{1}{2}s + \frac{1}{2})} f^{s+1}$, the last value becomes $\rho = 1$; writing for shortness $a^2 + \dots + c^2 + e^2 = \kappa^2$, we have

$$\begin{aligned} V &= \int \frac{dx \dots dz dw}{[(a-x)^2 + \dots + (c-z)^2 + (e-w)^2]^{\frac{1}{2}(s-1)}} \text{ over the } (s+1)\text{-dim. sphere } x^2 + \dots + z^2 + w^2 = f^2, \\ &= \frac{(\Gamma(\frac{1}{2})^{s+1})}{\Gamma(\frac{1}{2}s + \frac{1}{2})} \cdot \frac{f^{s+1}}{\kappa^{s-1}}, \quad \text{for an exterior point } \kappa > f, \\ &= \frac{(\Gamma(\frac{1}{2})^{s+1})}{\Gamma(\frac{1}{2}s + \frac{1}{2})} \left\{ (\frac{1}{2}s + \frac{1}{2})f^2 - (\frac{1}{2}s - \frac{1}{2})\kappa^2 \right\}, \quad \text{for an interior point } \kappa < f. \end{aligned}$$

134. The case of the ellipsoid $\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2} = 1$ for $s+1$ -dimensional space may be worked out by the theorem; this is, in fact, what is done in tridimensional space by Lejeune-Dirichlet in his Memoir of 1846 above referred to (p. 321).

Annex VIII. Prepotentials of the Homaloids. Art. Nos. 135 to 137.

135. We have in tridimensional space the series of figures—the plane, the line, the point; and there is in like manner in $(s+1)$ -dimensional space a corresponding series of $(s+1)$ terms; the s -coordinal plane, ... the $(s-1)$ -coordinal plane, ... the line, the point: say these are the homaloids or homaloidal figures. And, taking the density as uniform, or, what is the same thing, $= 1$, we may consider the prepotentials of these several figures in regard to an attracted point, which, for greater simplicity, is taken not to be on the figure.

136. The integral may be written

$$V = \int \frac{dw \dots dt}{\{(a-x)^2 + \dots + (c-z)^2 + (d-w)^2 + \dots + (e-t)^2 + u^2\}^{\frac{1}{2}s+q}},$$

which still relates to a $(s+1)$ -dimensional space: the $(s+1)$ coordinates of the attracted point are $(a, \dots, c, d, \dots, e, u)$, instead of being $(a, \dots, c, e)$; viz. we have the

[p. 409] s' coordinates $(a, \dots, c)$, the $s-s'$ coordinates $(d, \dots, e)$, and the $(s+1)$ th coordinate u : and the integration is extended over the $(s-s')$ -dimensional figure $w = -\infty$ to $+\infty, \dots, t = -\infty$ to $+\infty$. And it is also assumed that q is positive.

It is at once clear that we may reduce the integral to

$$V = \int \frac{dw \dots dt}{\{(a-x)^2 + \dots + (c-z)^2 + u^2 + w^2 + \dots + t^2\}^{\frac{1}{2}s+q}},$$

say for shortness

$$= \int \frac{dw \dots dt}{(A^2 + w^2 + \dots + t^2)^{\frac{1}{2}s+q}},$$

where $A^2 = (a - x)^2 + \dots + (c - z)^2 + u^2$, is a constant as regards the integration, and where the limits in regard to each of the $s - s'$ variables are $-\infty, +\infty$.

We may for these variables write $r\xi, \dots, r\zeta$, where $\xi^2 + \dots + \zeta^2 = 1$; and we then have $dw \dots dt = r^{s-s'-1} dr dS$, where dS is the element of surface of the $(s - s')$ -coordinal unit-sphere $\xi^2 + \dots + \zeta^2 = 1$. We thus obtain

$$V = \int \frac{r^{s-s'-1} dr}{(A^2 + r^2)^{\frac{1}{2}s+q}} \int dS,$$

where the integral in regard to r is taken from 0 to ∞ , and the integral $\int dS$ over the surface of the unit-sphere; hence by Annex I. the value of this last factor is $\frac{2(\Gamma\frac{1}{2})^{s-s'}}{\Gamma\frac{1}{2}(s-s')}$. The integral represented by the first factor will be finite, provided $\frac{1}{2}(s - s')$ is positive; which is the case for any value whatever of s' , if only q be positive.

The first factor is an integral such as is considered in Annex II.; to find its value we have only to write $r = Ax$, and we thus find it to be

$$\frac{1}{(A^2)^{\frac{1}{2}s'+q}} \cdot \frac{1}{A} \int_0^\infty \frac{x^{s-s'-1} dx}{(1+x)^{\frac{1}{2}s+q}} = \frac{1}{A^{s'+2q}} \cdot \frac{1}{2} \frac{\Gamma\frac{1}{2}(s-s') \Gamma(\frac{1}{2}s' + q)}{\Gamma(\frac{1}{2}s + q)},$$

and we thus have

$$V = \frac{(\Gamma\frac{1}{2})^{s-s'} \Gamma(\frac{1}{2}s' + q)}{\Gamma(\frac{1}{2}s + q)} \cdot \frac{1}{\{(a-x)^2 + \dots + (c-z)^2 + u^2\}^{\frac{1}{2}s'+q}}.$$

137. As a verification, observe that the prepotential equation $\square V = 0$, that is,

$$\left(\frac{d^2}{da^2} + \dots + \frac{d^2}{dc^2} + \frac{d^2}{dd^2} + \dots + \frac{d^2}{de^2} + \frac{d^2}{du^2} + \frac{2q+1}{u} \frac{d}{du} \right) V = 0,$$

for a function V , which contains only the $s' + 1$ variables $(a, \dots, c, u)$, becomes

$$\left(\frac{d^2}{da^2} + \dots + \frac{d^2}{dc^2} + \frac{d^2}{du^2} + \frac{2q+1}{u} \frac{d}{du} \right) V = 0,$$

which is satisfied by V , a constant multiple of $\{(a-x)^2 + \dots + (c-z)^2 + u^2\}^{-\frac{1}{2}s'-q}$.

Annex IX. The Gauss–Jacobi Theory of Epispheric Integrals. Art. No. 138.

[p. 410] **138.** The formula obtained (Annex IV. No. 110) is proved only for positive values of m ; but writing therein $q = 0$, $m = -\frac{1}{2}$, it becomes

$$\begin{aligned} & \int \sqrt{1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2}} \frac{dx \dots dz}{\{(a-x)^2 + \dots + (c-z)^2 + e^2\}^{\frac{1}{2}s}} \\ &= \frac{(\Gamma\frac{1}{2})^s}{\Gamma\frac{1}{2}s} f \dots h \int_0^\infty t^{-\frac{1}{2}} \left(1 - \frac{a^2}{t+f^2} - \dots - \frac{c^2}{t+h^2} - \frac{e^2}{t} \right)^{-\frac{1}{2}} \{(t+f^2) \dots (t+h^2)\}^{-\frac{1}{2}}, \end{aligned}$$

a formula which is obtainable as a particular case of the more general formula

$$V = \int \frac{dS}{\{(*x, \dots, z, w)^2\}^{\frac{1}{2}s}} = \frac{2(\Gamma\frac{1}{2})^s}{\Gamma(\frac{1}{2}s)} \int_{-A}^\infty \frac{dt}{\sqrt{-\text{Disct.} \dots [(*)X, \dots, Z, W, T)^2 + t(X^2 + \dots + Z^2 + W^2 - T^2)]}},$$

(notation to be presently explained), being a result obtained by Jacobi by a process which is in fact the extension to any number of variables of that used by Gauss¹ in his Memoir “Determinatio attractionis quam exercent planeta, &c.” (1818). I proceed to develop this theory.

139. Jacobi’s process has reference to a class of s -tuple integrals (including some of those here previously considered) which may be termed “epispheric”: viz. considering the $(s + 1)$ variables $(x, \dots, z, w)$ connected by the equation $x^2 + \dots + z^2 + w^2 = 1$, or say they are the coordinates of a point on a $(s + 1)$ -tuple unit-sphere, then the form is $\int U dS$, where dS is the element of the surface of the unit-sphere, and U is any function of the $s + 1$ coordinates; the integral is taken to be of the form $\int \frac{dS}{\{(*x, \dots, z, w, 1)^2\}^{\frac{1}{2}s}}$, and we then obtain the general result above referred to.

Before going further it is convenient to remark that, taking as independent variables the s coordinates $x, \dots, z$, we have $dS = \frac{dx \dots dz}{w}$, where w stands for $\pm \sqrt{1 - x^2 - \dots - z^2}$; we must in obtaining the integral take account of the two values of w , and finally extend the integral to the values of $x, \dots, z$ which satisfy $x^2 + \dots + z^2 \leq 1$.

If, as is ultimately done, in place of $x, \dots, z$ we write $\frac{x}{f}, \dots, \frac{z}{h}$ respectively, then the value of $dS = \frac{1}{f \dots h} \cdot \frac{dx \dots dz}{w}$, where w now stands for $\pm \sqrt{1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2}}$; we must, in finding the value of the integral, take account of the two values of w , and finally extend the integral to the values of $x, \dots, z$ which satisfy $\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2} \leq 1$.

[p. 411] **140.** The determination of the integral depends upon formulae for the transformation of the spherical element dS , and of the quadric function $(x, y, \dots, z, w, 1)^2$.

First, as regards the spherical element dS ; let the $s + 1$ variables $x, y, \dots, z, w$ which satisfy $x^2 + y^2 + \dots + z^2 + w^2 = 1$ be regarded as functions of the s independent variables $\theta, \phi, \dots, \psi$; then we have

$$dS = \frac{1}{w} \begin{vmatrix} \frac{dx}{d\theta} & \frac{dy}{d\theta} & \dots & \frac{dz}{d\theta} & \frac{dw}{d\theta} \\ \frac{dx}{d\phi} & \frac{dy}{d\phi} & \dots & \frac{dz}{d\phi} & \frac{dw}{d\phi} \\ \vdots & \vdots & & \vdots & \vdots \\ \frac{dx}{d\psi} & \frac{dy}{d\psi} & \dots & \frac{dz}{d\psi} & \frac{dw}{d\psi} \end{vmatrix} d\theta d\phi \dots d\psi = \frac{\partial(x, y, \dots, z, w)}{\partial(\theta, \phi, \dots, \psi, *)} d\theta d\phi \dots d\psi, \text{ for shortness.}$$

Suppose we effect on the $s + 1$ variables $(x, y, \dots, z, w)$ a transformation

$$x, y, \dots, z, w = \frac{X}{T}, \frac{Y}{T}, \dots, \frac{Z}{T}, \frac{W}{T},$$

thus introducing for the moment $s + 2$ variables $X, Y, \dots, Z, W, T$, which satisfy identically $X^2 + Y^2 + \dots + Z^2 + W^2 - T^2 = 0$; then, considering these as functions of the foregoing s independent variables $\theta, \phi, \dots, \psi$, we have

$$dS = \frac{1}{2^{s+1}} \frac{W}{T^{s+2}} \cdot \frac{\partial(X, Y, \dots, Z, W)}{\partial(\theta, \phi, \dots, \psi, *)} d\theta d\phi \dots d\psi.$$

141. Considering next the $s + 2$ variables $X, Y, \dots, Z, W, T$ as linear functions (with constant terms) of the $s + 1$ new variables $\xi, \eta, \dots, \zeta, \omega$, or say as linear functions of the $s + 2$ quantities $\xi, \eta, \dots, \zeta, \omega, 1$: which implies between them a linear relation

$$aX + bY + \dots + cZ + dW + eT = 1;$$

¹[*Ges. Werke*, t. iii, pp. 331–355.]

and assuming that we have identically

$$X^2 + Y^2 + \cdots + Z^2 + W^2 - T^2 = \xi^2 + \eta^2 + \cdots + \zeta^2 + \omega^2 - 1,$$

so that, in consequence of the left-hand side being $= 0$, the right-hand side is also $= 0$; viz. $\xi, \eta, \dots, \zeta, \omega$ are connected by

$$\xi^2 + \eta^2 + \cdots + \zeta^2 + \omega^2 = 1 :$$

[p. 412] let $d\Sigma$ represent the spherical element belonging to the coordinates $\xi, \eta, \dots, \zeta, \omega$. Considering these as functions of the foregoing s independent variables $\theta, \phi, \dots, \psi$, we have

$$d\Sigma = \frac{1}{\omega} \begin{vmatrix} \frac{d\xi}{d\theta} & \frac{d\eta}{d\theta} & \cdots & \frac{d\zeta}{d\theta} & \frac{d\omega}{d\theta} \\ \frac{d\xi}{d\phi} & \frac{d\eta}{d\phi} & \cdots & \frac{d\zeta}{d\phi} & \frac{d\omega}{d\phi} \\ \vdots & \vdots & & \vdots & \vdots \\ \frac{d\xi}{d\psi} & \frac{d\eta}{d\psi} & \cdots & \frac{d\zeta}{d\psi} & \frac{d\omega}{d\psi} \end{vmatrix} d\theta d\phi \dots d\psi = \frac{\partial(\xi, \eta, \dots, \zeta, \omega)}{\partial(\theta, \phi, \dots, \psi, *)} d\theta d\phi \dots d\psi.$$

142. In this expression we have $\xi, \eta, \dots, \zeta, \omega$, each of them a linear function of the $s + 2$ quantities $X, Y, \dots, Z, W, T$; the determinant is consequently a linear function of $s + 2$ like determinants obtained by substituting for the variables any $s + 1$ out of the $s + 2$ variables $X, Y, \dots, Z, W, T$; but in virtue of the equation

$$X^2 + Y^2 + \cdots + Z^2 + W^2 - T^2 = 0,$$

these $s + 2$ determinants are proportional to the quantities $X, Y, \dots, Z, W, T$ respectively, and the determinant thus assumes the form

$$\frac{aX + bY + \cdots + cZ + dW + eT}{T} \Delta,$$

where Δ is the like determinant with $(X, Y, \dots, Z, W)$, and where the coefficients $a, b, \dots, c, d, e$ are precisely those of the linear relation $aX + bY + \cdots + cZ + dW + eT = 1$; the last-mentioned expression is thus $\frac{1}{T}\Delta$, or, substituting for Δ its value, we have

$$d\Sigma = \frac{1}{T} \frac{\partial(X, Y, \dots, Z, W)}{\partial(\theta, \phi, \dots, \psi, *)} d\theta d\phi \dots d\psi;$$

viz. comparing with the foregoing expression for dS we have

$$dS = \frac{1}{T^{s+1}} d\Sigma,$$

which is the requisite formula for the transformation of dS .

143. Consider the integral

$$\int \frac{dS}{\{(*x, y, \dots, z, w, 1)^2\}^{\frac{1}{2}s}},$$

which, from its containing a single quadric function, may be called “one-quadric.” Then effecting the foregoing transformation,

$$x, y, \dots, z, w = \frac{X}{T}, \frac{Y}{T}, \dots, \frac{Z}{T}, \frac{W}{T},$$

[p. 413] and observing that

$$(*x, y, \dots, z, w, 1)^2 = \frac{1}{T^2} (*X, Y, \dots, Z, W, T)^2,$$

the integral becomes

$$= \int \frac{d\Sigma}{\{(*X, Y, \dots, Z, W, T)^2\}^{\frac{1}{2}s}},$$

where $X, Y, \dots, Z, W, T$ denote given linear functions (with constant coefficients) of the $s + 1$ variables $\xi, \eta, \dots, \zeta, \omega$, or, what is the same thing, given linear functions of the $s + 2$ quantities $\xi, \eta, \dots, \zeta, \omega, 1$, such that identically

$$X^2 + Y^2 + \dots + Z^2 + W^2 - T^2 = \xi^2 + \eta^2 + \dots + \zeta^2 + \omega^2 - 1.$$

We have then $\xi^2 + \eta^2 + \dots + \zeta^2 + \omega^2 - 1 = 0$, and $d\Sigma$ as the corresponding spherical element.

144. We may have $X, Y, \dots, Z, W, T$ such linear functions of $\xi, \eta, \dots, \zeta, \omega, 1$ that not only

$$X^2 + Y^2 + \dots + Z^2 + W^2 - T^2 = \xi^2 + \eta^2 + \dots + \zeta^2 + \omega^2 - 1$$

as above, but also

$$(*X, Y, \dots, Z, W, T)^2 = A\xi^2 + B\eta^2 + \dots + C\zeta^2 + E\omega^2 - L;$$

this being so, the integral becomes

$$\int \frac{d\Sigma}{[A\xi^2 + B\eta^2 + \dots + C\zeta^2 + E\omega^2 - L]^{\frac{1}{2}s}},$$

where the $s + 2$ coefficients $A, B, \dots, C, E, L$ are given by means of the identity

$$-(\theta + A)(\theta + B) \dots (\theta + C)(\theta + E)(\theta + L) = \text{Disct.} [(*X, Y, \dots, Z, W, T)^2 + \theta(X^2 + Y^2 + \dots + Z^2 + W^2 - T^2)];$$

viz. equating the discriminant to zero, we have an equation in θ , the roots whereof are $-A, -B, \dots, -C, -E, -L$.

The integral is

$$\int \frac{d\Sigma}{[(A - L)\xi^2 + (B - L)\eta^2 + \dots + (C - L)\zeta^2 + (E - L)\omega^2]^{\frac{1}{2}s}}$$

which is of the form

$$\int \frac{d\Sigma}{[a\xi^2 + b\eta^2 + \dots + c\zeta^2 + e\omega^2]^{\frac{1}{2}s}},$$

where I provisionally assume that $a, b, \dots, c, e$ are all positive.

145. To transform this, in place of the $s + 1$ variables $\xi, \eta, \dots, \zeta, \omega$ connected by $\xi^2 + \eta^2 + \dots + \zeta^2 + \omega^2 = 1$, we introduce the $s + 1$ variables $x, y, \dots, z, w$, such that

$$x = \frac{\xi\sqrt{a}}{\rho}, \quad y = \frac{\eta\sqrt{b}}{\rho}, \quad \dots, \quad z = \frac{\zeta\sqrt{c}}{\rho}, \quad w = \frac{\omega\sqrt{e}}{\rho},$$

[p. 414] where

$$\rho^2 = a\xi^2 + b\eta^2 + \dots + c\zeta^2 + e\omega^2,$$

and consequently

$$x^2 + y^2 + \dots + z^2 + w^2 = 1.$$

Hence, writing dS to denote the spherical element corresponding to the point $(x, y, \dots, z, w)$, we have, by a former formula,

$$dS = \frac{1}{w} \frac{\partial(x, y, \dots, z, \omega\sqrt{e}/\rho)}{\partial(\theta, \phi, \dots, \psi, *)} d\theta d\phi \dots d\psi = \frac{(ab \dots ce)^{\frac{1}{2}}}{\rho^{s+1}} d\Sigma,$$

or, what is the same thing,

$$d\Sigma = \frac{\rho^{s+1}}{(ab \dots ce)^{\frac{1}{2}}} dS.$$

Hence, integrating each side, and observing that $\int dS$, taken over the whole spherical surface $x^2 + y^2 + \dots + z^2 + w^2 = 1$, is $= \frac{2(\Gamma(\frac{1}{2})^{s+1}}{\Gamma(\frac{1}{2}s + \frac{1}{2})}$, we have

$$\int \frac{d\Sigma}{[a\xi^2 + b\eta^2 + \dots + c\zeta^2 + e\omega^2]^{\frac{1}{2}s + \frac{1}{2}}} = \frac{2(\Gamma(\frac{1}{2})^{s+1}}{\Gamma(\frac{1}{2}s + \frac{1}{2})} \cdot \frac{1}{(ab \dots ce)^{\frac{1}{2}}}.$$

146. For $a, b, \dots, c, e$ write herein $a + \theta, b + \theta, \dots, c + \theta, e + \theta$ respectively, and multiply each side by θ^{q-1} , where q is any positive integer or fractional number less than $\frac{1}{2}s$: integrate from $\theta = 0$ to $\theta = \infty$. On the left-hand side, attending to the relation $\xi^2 + \eta^2 + \dots + \zeta^2 + \omega^2 = 1$, the integral in regard to θ is

$$\int_0^\infty \frac{\theta^{q-1} d\theta}{(\rho^2 + \theta)^{\frac{1}{2}(s+1)}},$$

where $\rho^2 = a\xi^2 + b\eta^2 + \dots + c\zeta^2 + e\omega^2$, is independent of θ as before; the value of the definite integral is

$$\frac{\Gamma\{\frac{1}{2}(s+1) - q\} \Gamma(q)}{\Gamma(\frac{1}{2}(s+1))} \cdot \frac{1}{\rho^{s+1-2q}},$$

which, replacing ρ by its value and multiplying by $d\Sigma$, and prefixing the integral sign, gives the left-hand side; hence, forming the equation and dividing by a numerical factor, we have

$$\int \frac{d\Sigma}{[a\xi^2 + \dots + c\zeta^2 + e\omega^2]^{\frac{1}{2}s + \frac{1}{2} - q}} = \frac{2(\Gamma(\frac{1}{2})^{s+1}}{\Gamma(\frac{1}{2}(s+1) - q) \Gamma(q)} \cdot \frac{1}{(ab \dots ce)^{\frac{1}{2}}} \int_0^\infty dt t^{q-1} [(t+a) \dots (t+c)(t+e)]^{-\frac{1}{2}}.$$

In particular, if $q = \frac{1}{2}$, then

$$\int \frac{d\Sigma}{[a\xi^2 + \dots + c\zeta^2 + e\omega^2]^{\frac{1}{2}s}} = \frac{2(\Gamma(\frac{1}{2})^s}{\Gamma(\frac{1}{2}s)} \int_0^\infty dt [(t+a) \dots (t+c)(t+e)]^{-\frac{1}{2}};$$

[p. 415] or, if for $a, \dots, c, e$ we restore the values $A - L, \dots, C - L, E - L$, then

$$\int \frac{d\Sigma}{[(A-L)\xi^2 + \dots + (C-L)\zeta^2 + (E-L)\omega^2]^{\frac{1}{2}s}} = \frac{2(\Gamma(\frac{1}{2})^s}{\Gamma(\frac{1}{2}s)} \int_{-L}^\infty dt [(t+A) \dots (t+C)(t+E)(t+L)]^{-\frac{1}{2}},$$

viz. we thus have

$$\int \frac{dS}{[(*)x, \dots, z, w, 1]^2]^{\frac{1}{2}s}} = \frac{2(\Gamma(\frac{1}{2})^s}{\Gamma(\frac{1}{2}s)} \int_{-L}^\infty dt [(t+A) \dots (t+C)(t+E)(t+L)]^{-\frac{1}{2}},$$

where $(t+A) \dots (t+C)(t+E)(t+L)$ is in fact a given rational and integral function of t ; viz. it is

$$= -\text{Disct.} \dots [(*)X, \dots, Z, W, T)^2 + t(X^2 + \dots + Z^2 + W^2 - T^2)].$$

147. Consider, in particular, the integral

$$\int \frac{dS}{[(a-fx)^2 + \dots + (c-hz)^2 + (e-kw)^2 + l^2]^{\frac{1}{2}s}};$$

here

$$\begin{aligned} & (aT - fX)^2 + \dots + (cT - hZ)^2 + (eT - kW)^2 + l^2 T^2 + t(X^2 + \dots + Z^2 + W^2 - T^2) \\ &= (f^2 + t)X^2 + \dots + (h^2 + t)Z^2 + (k^2 + t)W^2 + (a^2 + \dots + c^2 + e^2 + l^2 - t)T^2 \\ & \quad - 2afXT - \dots - 2chZT - 2ekWT; \end{aligned}$$

viz. the discriminant taken negatively is

$$\begin{vmatrix} t + f^2 & \dots & \dots & -af \\ \vdots & \ddots & & \vdots \\ \dots & & t + k^2 & -ek \\ -af & \dots & -ek & -(a^2 + \dots + c^2 + e^2 + l^2) + t \end{vmatrix},$$

which is

$$= (t + A) \dots (t + C)(t + E)(t + L);$$

and consequently $-A, \dots, -C, -E, -L$ are the roots of the equation

$$1 - \frac{a^2}{t + f^2} - \dots - \frac{c^2}{t + h^2} - \frac{e^2}{t + k^2} - \frac{l^2}{t} = 0.$$

148. The roots are all real; moreover there is one and only one positive root. Hence, taking $-L$ to be the positive root, we have $A, \dots, C, E, L$ all positive, and therefore *à fortiori* $A - L, \dots, C - L, E - L$ all positive: which agrees with a foregoing

[p. 416] provisional assumption. Or, writing for greater convenience θ to denote the positive quantity $-L$, that is, taking θ to be the positive root of the equation

$$1 - \frac{a^2}{\theta + f^2} - \dots - \frac{c^2}{\theta + h^2} - \frac{e^2}{\theta + k^2} - \frac{l^2}{\theta} = 0,$$

we have

$$\int \frac{dS}{[(a-fx)^2 + \dots + (c-hz)^2 + (e-kw)^2 + l^2]^{\frac{1}{2}s}} = \frac{2(\Gamma(\frac{1}{2}))^s}{\Gamma(\frac{1}{2}s)} \int_{\theta}^{\infty} dt \left(1 - \frac{a^2}{t + f^2} - \dots - \frac{c^2}{t + h^2} - \frac{e^2}{t + k^2} - \frac{l^2}{t}\right)^{-\frac{1}{2}} [t$$

or, what is the same thing, we have

$$\frac{1}{f \dots h k} \int \frac{dx \dots dz}{w[(a-x)^2 + \dots + (c-z)^2 + (e-kw)^2 + l^2]^{\frac{1}{2}s}} = \frac{2(\Gamma(\frac{1}{2}))^s}{\Gamma(\frac{1}{2}s)} \int_{\theta}^{\infty} dt \left(1 - \frac{a^2}{t + f^2} - \dots - \frac{c^2}{t + h^2} - \frac{e^2}{t + k^2} - \frac{l^2}{t}\right)^{-\frac{1}{2}} [t$$

where on the left-hand side w now denotes $\sqrt{1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2}}$, and the limiting equation is

$$\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2} = 1.$$

149. Suppose $l = 0$: then, if

$$\frac{a^2}{f^2} + \dots + \frac{c^2}{h^2} + \frac{e^2}{k^2} > 1,$$

the equation

$$1 - \frac{a^2}{\theta + f^2} - \cdots - \frac{c^2}{\theta + h^2} - \frac{e^2}{\theta + k^2} = 0$$

has a positive root differing from zero, which may be represented by the same letter θ ; but if

$$\frac{a^2}{f^2} + \cdots + \frac{c^2}{h^2} + \frac{e^2}{k^2} < 1,$$

then the positive root of the original equation becomes $= 0$; viz. as l gradually diminishes to zero, the positive root also diminishes and becomes ultimately zero.

Hence, writing $l = 0$, we have

$$\int \frac{dS}{[(a - fx)^2 + \cdots + (c - hz)^2 + (e - kw)^2]^{\frac{1}{2}s}},$$

or, what is the same thing,

$$\frac{1}{f \cdots h k} \int \frac{dx \cdots dz}{w[(a - x)^2 + \cdots + (c - z)^2 + (e - kw)^2]^{\frac{1}{2}s}},$$

[p. 417] θ now denoting either the positive root of the equation

$$1 - \frac{a^2}{\theta + f^2} - \cdots - \frac{c^2}{\theta + h^2} - \frac{e^2}{\theta + k^2} = 0,$$

or else 0, according as

$$\frac{a^2}{f^2} + \cdots + \frac{c^2}{h^2} + \frac{e^2}{k^2} > 1 \text{ or } < 1.$$

In the case $\frac{a^2}{f^2} + \cdots + \frac{c^2}{h^2} + \frac{e^2}{k^2} < 1$, the inferior limit being then 0, this is, in fact, Jacobi's theorem (*Crelle*, t. xii. p. 69, 1834); but Jacobi does not consider the general case where l is not $= 0$, nor does he give explicitly the formula in the other case

$$l = 0, \quad \frac{a^2}{f^2} + \cdots + \frac{c^2}{h^2} + \frac{e^2}{k^2} > 1.$$

150. Suppose $k = 0$, e being in the first instance not $= 0$: then the former alternative holds good; and observing, in regard to the form which contains w in the denominator, that we can now take account of the two values by simply multiplying by 2, we have

$$\int \frac{dS}{[(a - fx)^2 + \cdots + (c - hz)^2 + e^2]^{\frac{1}{2}s}} = \frac{2}{f \cdots h} \int \frac{dx \cdots dz}{[(a - x)^2 + \cdots + (c - z)^2 + e^2]^{\frac{1}{2}s}}$$

(where on the right-hand side w denotes $\sqrt{1 - \frac{x^2}{f^2} - \cdots - \frac{z^2}{h^2}}$, and the limiting equation being

$$\frac{x^2}{f^2} + \cdots + \frac{z^2}{h^2} = 1); \text{ each } = \frac{2(\Gamma(\frac{1}{2}))^s}{\Gamma(\frac{1}{2}s)} \int_{\theta}^{\infty} dt \left(1 - \frac{a^2}{t + f^2} - \cdots - \frac{c^2}{t + h^2} - \frac{e^2}{t} \right)^{-\frac{1}{2}} [t(t + f^2) \cdots (t + h^2)]^{-\frac{1}{2}},$$

where θ is here the positive root of the equation $1 - \frac{a^2}{\theta + f^2} - \cdots - \frac{c^2}{\theta + h^2} - \frac{e^2}{\theta} = 0$, which is the formula referred to at the beginning of the present Annex. We may in the formula write $e = 0$, thus obtaining the theorem under two different forms for the cases

$$\frac{a^2}{f^2} + \cdots + \frac{c^2}{h^2} > 1 \text{ and } < 1 \text{ respectively.}$$

Annex X. Methods of Lejeune-Dirichlet and Boole. Art. Nos. 151 to 162.

151. The notion, that the density ρ is a discontinuous function vanishing for points outside the attracting mass, has been made use of in a different manner by Lejeune-Dirichlet (1839) and Boole (1857): viz. supposing that ρ has a given value $= f(x, \dots, z)$ within a given closed surface S and is $= 0$ outside the surface, these geometers in the expression of a potential or prepotential integral replace ρ by a definite integral which possesses the discontinuity in question, viz. it is $= f(x, \dots, z)$ for points inside

[p. 418] the surface and $= 0$ for points outside the surface; and then in the potential or prepotential integral they extend the integration over the whole of infinite space, thus getting rid of the equation of the surface as a limiting equation for the multiple integral.

152. Lejeune-Dirichlet's paper "Sur une nouvelle méthode pour la détermination des intégrales multiples" is published in *Comptes Rendus*, t. viii. pp. 155–160 (1839), and *Liouville*, t. iv. pp. 164–168 (same year). The process is applied to the form

$$-\frac{1}{p-1} \frac{d}{da} \int \frac{dx dy dz}{\{(a-x)^2 + (b-y)^2 + (c-z)^2\}^{\frac{1}{2}(p-1)}},$$

taken over the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$; but it would be equally applicable to the triple integral itself, or say to the s -tuple integral

$$\int \frac{dx \dots dz}{\{(a-x)^2 + \dots + (c-z)^2\}^{\frac{1}{2}s+q}}$$

or, indeed, to

$$\int \frac{dx \dots dz}{\{(a-x)^2 + \dots + (c-z)^2 + e^2\}^{\frac{1}{2}s+q}}$$

taken over the ellipsoid $\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2} = 1$; but it may be as well to attend to the first form, as more resembling that considered by the author.

153. Since $\frac{2}{\pi} \int_0^\infty \frac{\sin \lambda \phi}{\phi} \cos \lambda \phi d\phi$ is $= 1$ or 0 , according as λ is < 1 or > 1 , it follows that the integral is equal to the real part of the following expression,

$$\frac{2}{\pi} \int_0^\infty \frac{d\phi}{\phi} e^{i\phi} \int \frac{e^{-i\phi(x^2+\dots+z^2)} dx \dots dz}{\{(a-x)^2 + \dots + (c-z)^2\}^{\frac{1}{2}s+q}},$$

where the integrations in regard to $x, \dots, z$ are now to be extended from $-\infty$ to $+\infty$ for each variable. A further transformation is necessary: since

$$\int_0^\infty e^{-x\tau} \tau^{r-1} d\tau = \frac{\Gamma(r)}{x^r} \quad (x \text{ positive, and } r \text{ positive and } < 1),$$

writing herein $(a-x)^2 + \dots + (c-z)^2$ for x , and $\frac{1}{2}s+q$ for r , we have

$$\frac{1}{\{(a-x)^2 + \dots + (c-z)^2\}^{\frac{1}{2}s+q}} = \frac{1}{\Gamma(\frac{1}{2}s+q)} \int_0^\infty d\tau \tau^{\frac{1}{2}s+q-1} e^{-\tau\{(a-x)^2+\dots+(c-z)^2\}},$$

and the value is thus

$$\frac{2}{\pi \Gamma(\frac{1}{2}s+q)} \int_0^\infty \frac{d\phi}{\phi} e^{i\phi} \int_0^\infty d\tau \tau^{\frac{1}{2}s+q-1} \int e^{-i\phi(x^2+\dots+z^2)} e^{-\tau\{(a-x)^2+\dots+(c-z)^2\}} dx \dots dz,$$

[p. 419] where the integral in regard to the variables $(x, \dots, z)$ is

$$= e^{-i\phi\tau \frac{a^2+\dots+c^2}{\tau+i\phi}} \int dx e^{-(\tau+i\phi)\left(x-\frac{\tau a}{\tau+i\phi}\right)^2} \dots,$$

and here the x -integral is

$$= \pi^{\frac{1}{2}} \sqrt{\frac{1}{\tau+i\phi}} = \pi^{\frac{1}{2}} \frac{e^{-\frac{1}{2}i\theta}}{\sqrt{\tau^2+\phi^2}},$$

and the like for the other integrals up to the z -integral. The resulting value is thus

$$= \pi^{\frac{s}{2}} (f \dots h) e^{-i\phi\tau \frac{a^2+\dots+c^2}{\tau+i\phi}} e^{-\frac{1}{2}is\theta} \frac{1}{(\tau^2+\phi^2)^{\frac{1}{4}s}},$$

viz. writing $\psi = \frac{\phi}{\tau}$, $d\phi = \frac{\phi}{\tau} dt = \frac{\psi}{\tau} dt$, we have

$$= \frac{2\pi^{\frac{1}{2}s-1}}{\Gamma(\frac{1}{2}s+q)} (f \dots h) \int_0^\infty dt \frac{t^{q-1}}{\sqrt{(t^2+\tau^2)} \dots (h^2+t)} \int_0^\infty d\phi \phi^{q+r-1} e^{i\phi\sigma} \sin \phi d\phi.$$

154. But we have to consider only the real part of this expression; viz. writing for shortness $\sigma = \frac{a^2+\dots+c^2}{t+f^2} + \dots + \frac{c^2}{t+h^2}$, we require the real part of

$$e^{-i\sigma\phi} \int_0^\infty e^{i\sigma\phi} \phi^{q-1} \sin \phi d\phi.$$

Writing here for $\sin \phi$ its exponential value $\frac{1}{2i}(e^{i\phi} - e^{-i\phi})$, and using the formula

$$\frac{1}{(1-\sigma)^q} = \frac{1}{\Gamma(q)} \int_0^\infty d\phi \phi^{q-1} e^{-(1-\sigma)\phi} \quad (\sigma \text{ positive}),$$

and the like one

$$\frac{1}{(\sigma-1)^q} = \frac{1}{\Gamma(q)} e^{i\pi q} \int_0^\infty d\phi \phi^{q-1} e^{-(\sigma-1)\phi} \quad (\sigma \text{ negative}),$$

(in which formulae q must be positive and less than 1), we see that the real part in question is 0, or is

$$\frac{\Gamma q \sin(q+1)\pi}{2(1-\sigma)^q} = \frac{\pi}{2\Gamma(1-q)} \cdot \frac{1}{(1-\sigma)^q},$$

according as $\sigma > 1$ or $\sigma < 1$.

155. If the point is interior, $\frac{a^2}{f^2} + \dots + \frac{c^2}{h^2} < 1$, and consequently also $\sigma < 1$, and the value, writing $(\Gamma \frac{1}{2})^s$ instead of $\pi^{\frac{1}{2}s}$, is

[p. 420] But if the point be exterior, $\frac{a^2}{f^2} + \dots + \frac{c^2}{h^2} > 1$, and hence, writing θ for the positive root of the equation, $\sigma = 1$; viz. θ is the positive root of the equation $\frac{a^2}{\theta+f^2} + \dots + \frac{c^2}{\theta+h^2} = 1$; then $t = 0$, σ is greater than 1, and continues so as t increases, until, for $t = \theta$, σ becomes = 1, and for larger values of t we have $\sigma < 1$; and the expression thus is

$$= \frac{(\Gamma \frac{1}{2})^s}{\Gamma(\frac{1}{2}s+q)\Gamma(1-q)} (f \dots h) \int_\theta^\infty dt t^{q-1} [(t+f^2) \dots (t+h^2)]^{-\frac{1}{2}} \left(1 - \frac{a^2}{t+f^2} - \dots - \frac{c^2}{t+h^2}\right)^{-q};$$

viz. the two expressions, in the cases of an interior point and an exterior point respectively, give the value of the integral

$$\int \frac{dx \dots dz}{\{(a-x)^2 + \dots + (c-z)^2\}^{\frac{1}{2}s+q}}.$$

This is, in fact, the formula of Annex IV. No. 110, writing therein $e = 0$ and $m = -q$.

156. Boole's researches are contained in two memoirs dated 1846, "On the Analysis of Discontinuous Functions," *Trans. Royal Irish Academy*, vol. xxi. (1848), pp. 124–139, and "On a certain Multiple Definite Integral," do. pp. 140–150 (the particular theorem about to be referred to is stated in the postscript of this memoir), and in the memoir "On the Comparison of Transcendents, with certain applications to the theory of Definite Integrals," *Phil. Trans.*, vol. cxlvii. (1857), pp. 745–803, the theorem being the third example, p. 794. The method is similar to, and was in fact suggested by, that of Lejeune-Dirichlet; the auxiliary theorem made use of in the memoir of 1857 for the representation of the discontinuity being

$$\phi(\sigma) = \frac{1}{\pi \Gamma(-q)} \int_{-\infty}^{\infty} \int_0^{\infty} du dv v^q \cos\{(u - \sigma)v + \frac{1}{2}q\pi\} \phi(u),$$

which is a deduction from Fourier's theorem.

Changing the notation (and in particular writing s and $\frac{1}{2}s + q$ for his n and i), the method is here applied to the determination of the s -tuple integral

$$V = \int dx \dots dz \frac{\phi\left(\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2}\right)}{\{(a-x)^2 + \dots + (c-z)^2 + e^2\}^{\frac{1}{2}s+q}},$$

where ϕ is an arbitrary function, taken over the ellipsoid $\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2} = 1$.

157. The process is as follows: we have

$$\frac{\phi\left(\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2}\right)}{\{(a-x)^2 + \dots + (c-z)^2 + e^2\}^{\frac{1}{2}s+q}} = \frac{1}{\pi \Gamma(\frac{1}{2}s+q)} \int_0^{\infty} \int_0^{\infty} \int_0^{\infty} du dv d\tau v^{\frac{1}{2}s+q} \tau^{\frac{1}{2}s+q-1} e^{-\tau v} \cos\left\{\left(u - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2}\right)v + \frac{1}{2}(\frac{1}{2}s+q)\pi\right\}.$$

[p. 421] viz. the right-hand side is here equal to the left-hand side or is $= 0$, according as $\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2} < 1$ or > 1 . V is consequently obtained by multiplying the right-hand side by $dx \dots dz$ and integrating from $-\infty$ to $+\infty$ for each variable.

Hence, changing the order of the integration,

$$V = \frac{1}{\pi \Gamma(\frac{1}{2}s+q)} \int_0^{\infty} \int_0^{\infty} \int_0^{\infty} du dv d\tau v^{\frac{1}{2}s+q} \tau^{\frac{1}{2}s+q-1} e^{-\tau v} \phi u \cdot \Pi,$$

where

$$\Pi = \int dx \dots dz \cos\left(uv - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2} + \tau\{(a-x)^2 + \dots + (c-z)^2\}v + \frac{1}{2}(\frac{1}{2}s+q)\pi\right).$$

Now if

$$\xi = \sqrt{\frac{v}{f^2}} x, \dots, \zeta = \sqrt{\frac{v}{h^2}} z.$$

158. Substituting, and integrating with respect to $\xi, \dots, \zeta$ between the limits $-\infty, +\infty$, we have

$$\Pi = \frac{\pi^{\frac{s}{2}} f \dots h}{\sqrt{(1+f^2\tau v) \dots (1+h^2\tau v)}} \cos\left\{\left(u - \frac{a^2\tau v}{1+f^2\tau v} - \dots - \frac{c^2\tau v}{1+h^2\tau v}\right)v + e^2\tau v + \frac{1}{2}q\pi\right\},$$

or, what is the same thing, writing $\frac{t}{v}$ in place of τ , this is

$$= \frac{\pi^{\frac{s}{2}} f \dots h}{\sqrt{(1+f^2t) \dots (1+h^2t)}} \cos \left\{ \left(u - \frac{a^2t}{1+f^2t} - \dots - \frac{c^2t}{1+h^2t} \right) v + e^2t + \frac{1}{2}q\pi \right\},$$

that is, writing

$$\sigma = \frac{a^2t}{1+f^2t} + \dots + \frac{c^2t}{1+h^2t} + e^2t,$$

we have

$$V = \frac{\pi^{\frac{s}{2}-1} (f \dots h)}{\Gamma(\frac{1}{2}s + q)} \int_0^\infty \int_0^\infty du dt t^{q-1} v^q \cos[(u - \sigma)v + \frac{1}{2}q\pi] \phi u$$

or, writing $\pi^{\frac{1}{2}s-1} = (\Gamma\frac{1}{2})^s \pi^{-1}$, this is

$$= \frac{(\Gamma\frac{1}{2})^s}{\pi \Gamma(\frac{1}{2}s + q)} (f \dots h) \int_0^\infty dt t^{q-1} [(1+f^2t) \dots (1+h^2t)]^{-\frac{1}{2}} \int_0^\infty \int_0^\infty du dv v^q \cos[(u - \sigma)v + \frac{1}{2}q\pi] \phi u.$$

159. Boole writes

$$\frac{1}{\pi} \int_0^\infty \int_0^\infty du dv v^q \cos\{(u - \sigma)v + \frac{1}{2}q\pi\} \phi u = \left(-\frac{d}{d\sigma}\right)^q \phi(\sigma);$$

viz. starting from Fourier's theorem,

$$\frac{1}{\pi} \iint du dv \cos(u - \sigma)v \cdot \phi u = \phi(\sigma),$$

[p. 422] where $\phi(\sigma)$ is regarded as vanishing except when σ is between the limits 0, 1, and the limits of u are taken to be 0, 1 accordingly, then, according to an admissible theory of general differentiation, we have the result in question. He has in the formula $\frac{1}{s}$ instead of my t ; and he proceeds, "Here σ increases continually with s . As s varies from 0 to ∞ , σ also varies from 0 to ∞ . To any positive limits of σ will correspond positive limits of s ; and these, as will hereafter appear"—this refers to his note "B"—"will in certain cases replace the limits 0 and ∞ in the expression for V ."

160. It seems better to deal with the result in the following manner, as in part shown p. 803 of Boole's memoir. Writing the integral in the form

$$V = \frac{(\Gamma\frac{1}{2})^s (f \dots h)}{\pi \Gamma(\frac{1}{2}s + q)} \int_0^\infty \int_0^1 du dt t^{q-1} \{(t + f^2) \dots (t + h^2)\}^{-\frac{1}{2}} \phi(u) \int_0^\infty dv v^q \cos[(u - \sigma)v + \frac{1}{2}q\pi],$$

effect the integration in regard to v ; viz. according as u is greater or less than σ , then

$$\begin{aligned} \int_0^\infty dv v^q \cos\{(u - \sigma)v + \frac{1}{2}q\pi\} &= \frac{\Gamma(q+1) \sin(q+1)\pi}{(u - \sigma)^{q+1}} \quad \text{or} \quad 0, \\ &= \frac{\pi}{\Gamma(-q) (u - \sigma)^{q+1}} \quad \text{or} \quad 0; \end{aligned}$$

and consequently, writing for σ its value,

$$V = \dots \left(u - \frac{a^2t}{1+f^2t} - \dots \right)^{-q-1} \phi(u), \quad \text{or } 0, \quad \text{as above.}$$

161. To further explain this, consider t as an x -coordinate and u as a y -coordinate; then, tracing the curve

$$y = \frac{a^2x}{f^2+x} + \cdots + \frac{c^2x}{h^2+x},$$

for positive values of x this is a mere hyperbolic branch, as shown in the figure, viz. $x = 0, y = 0$; and as x continually increases to ∞ , y continually decreases to zero.

The limits are originally taken to be from $u = 0$ to $u = 1$ and $t = 0$ to $t = \infty$, viz. over the infinite strip bounded by the lines $t0, 01, 11$; but within these limits the

[p. 423] function under the integral sign is to be replaced by zero whenever the values u, t are such that u is less than $\frac{a^2t}{f^2+t} + \cdots + \frac{c^2t}{h^2+t} + e^2t$, viz. when the values belong to a point in the shaded portion of the strip; the integral is therefore to be extended only over the unshaded portion of the strip; viz. the value is

$$V = \frac{(\Gamma(\frac{1}{2}))^s (f \dots h)}{\Gamma(-q) \Gamma(\frac{1}{2}s + q)} \iint dt du t^{q-1} \{(t+f^2) \dots (t+h^2)\}^{-\frac{1}{2}} \left(u - \frac{a^2t}{f^2+t} - \cdots - \frac{c^2t}{h^2+t} - e^2t\right)^{-q-1} \phi u,$$

the double integral being taken over the unshaded portion of the strip; or, what is the same thing, the integral in regard to u is to be taken from $u = \frac{a^2t}{f^2+t} + \cdots + \frac{c^2t}{h^2+t} + e^2t$ (say from $= \sigma$) to $u = 1$, and then the integral in regard to t is to be taken from $t = \theta$ to $t = \infty$, where, as before, θ is the positive root of the equation $\sigma = 1$, that is, of

$$\frac{a^2t}{f^2+t} + \cdots + \frac{c^2t}{h^2+t} + e^2t = 1.$$

162. Write $u = \sigma + (1 - \sigma)x$, and therefore $u - \sigma = (1 - \sigma)x$, $1 - u = (1 - \sigma)(1 - x)$, and $du = (1 - \sigma) dx$; then the limits $(1, 0)$ of x correspond to the limits $(1, \sigma)$ of u , and the formula becomes

$$V = \frac{(\Gamma(\frac{1}{2}))^s (f \dots h)}{\Gamma(-q) \Gamma(\frac{1}{2}s + q)} \int_{\theta}^{\infty} \int_0^1 dt dx t^{q-1} \{(t+f^2) \dots (t+h^2)\}^{-\frac{1}{2}} (1 - \sigma)^{-q} x^{-q-1} \phi[\sigma + (1 - \sigma)x],$$

where σ is retained in place of its value $\frac{a^2t}{f^2+t} + \cdots + \frac{c^2t}{h^2+t} + e^2t$. This is, in fact, a form (deduced from Boole's result in the memoir of 1846) given by me, *Cambridge and Dublin Mathematical Journal*, vol. ii. (1847), p. 219, [44].

If in particular $\phi u = (1 - u)^{q+m}$, then $\phi \sigma + (1 - \sigma)x = (1 - \sigma)^{q+m} (1 - x)^{q+m}$, and thence

$$\int_0^1 x^{-q-1} \phi[\sigma + (1 - \sigma)x] dx = (1 - \sigma)^{q+m} \int_0^1 x^{-q-1} (1 - x)^{q+m} dx = \frac{\Gamma(-q) \Gamma(1 + q + m)}{\Gamma(1 + m)},$$

and then, restoring for σ its value, we have

$$V = \frac{(\Gamma(\frac{1}{2}))^s \Gamma(1 + q + m)}{\Gamma(\frac{1}{2}s + q) \Gamma(1 + m)} (f \dots h) \int_{\theta}^{\infty} dt t^{q-1} \{(t+f^2) \dots (t+h^2)\}^{-\frac{1}{2}} \left(1 - \frac{a^2t}{f^2+t} - \cdots - \frac{c^2t}{h^2+t} - e^2t\right)^{q+m}$$

as the value of the integral

$$\int \frac{(1 - \frac{x^2}{f^2} - \cdots - \frac{z^2}{h^2})^m dx \dots dz}{\{(a-x)^2 + \cdots + (c-z)^2 + e^2\}^{\frac{1}{2}s+q}}$$

taken over the ellipsoid $\frac{x^2}{f^2} + \cdots + \frac{z^2}{h^2} = 1$. This is, in fact, the theorem of Annex IV. No. 110 in its general form; but the proof assumes that q is positive.

[End of Paper 607: A Memoir on Prepotentials.]

608. [Extract from a] Report on Mathematical Tables

[From the Report of the British Association for the Advancement of Science, (1873), pp. 3, 4.]

[p. 424] It was necessary as a preliminary to form a classification of mathematical (numerical) tables; and the following classification was drawn up by Prof. Cayley and adopted by the Committee.

A. Auxiliary for non-logarithmic computations.

1. Multiplication.
2. Quarter-squares.
3. Squares, cubes, and higher powers, and reciprocals.

B. Logarithmic and circular.

4. Logarithms (Briggian) and antilogarithms (do.); addition and subtraction logarithms, &c.
5. Circular functions (sines, cosines, &c.), natural, and lengths of circular arcs.
6. Circular functions (sines, cosines, &c.), logarithmic.

C. Exponential.

7. Hyperbolic logarithms.
8. Do. antilogarithms (e^x) and $h.1 \tan(45^\circ + \frac{1}{2}\phi)$, and hyperbolic sines, cosines, &c., natural and logarithmic.

D. Algebraic constants.

9. Accurate integer or fractional values. Bernoulli's Numbers, $\Delta^n 0^m$, &c. Binomial coefficients.
10. Decimal values auxiliary to the calculation of series.

[p. 425]

E. 11. Transcendental constants, e, π, γ , &c., and their powers and functions.

F. Arithmological.

12. Divisors and prime numbers. Prime roots. The Canon arithmeticus, &c.
13. The Pellian equation.
14. Partitions.
15. Quadratic forms $\pm a^2 \pm b^2$, &c., and partition of numbers into squares, cubes, and biquadrates.
16. Binary, ternary, &c. quadratic, and higher forms.
17. Complex theories.

G. Transcendental functions.

18. Elliptic.
19. Gamma.
20. Sine-integral, cosine-integral, and exponential-integral.

21. Bessel's and allied functions.
22. Planetary coefficients for given $\frac{a}{a'}$.
23. Logarithmic transcendental.
24. Miscellaneous.

Several of these classes need some little explanation. Thus D 9 and 10 are intended to include the same class of constants, the only difference being that in 9 accurate values are given, while in 10 they are only approximate; thus, for example, the accurate Bernoulli's numbers as vulgar fractions, and the decimal values of the same to (say) ten places are placed in different classes, as the former are of theoretical interest, while the latter are only of use in calculation. It is not necessary to enter into further detail with respect to the classification, as in point of fact it is only very partially followed in the Report.

609. On the Analytical Forms called Factions

[From the Report of the British Association for the Advancement of Science, (1875), p. 10.]

[p. 426] A FACTION is a product of differences such that each letter occurs the same number of times; thus we have a quadrifaction where each letter occurs twice, a cubifaction where each letter occurs three times, and so on. A broken faction is one which is a product of factions having no common letter; thus

$$(a - b)^2(c - d)(d - e)(e - c)$$

is a broken quadrifaction, the product of the quadrifactions

$$(a - b)^2 \quad \text{and} \quad (c - d)(d - e)(e - c).$$

We have, in regard to quadrifactions, the theorem that every quadrifaction is a sum of broken quadrifactions such that each component quadrifaction contains two or else three letters. Thus we have the identity

$$2(a - b)(b - c)(c - d)(d - a) = (b - c)^2 \cdot (a - d)^2 - (c - a)^2 \cdot (b - d)^2 + (a - b)^2 \cdot (c - d)^2,$$

which verifies the theorem in the case of a quadrifaction of four letters; but the verification even in the next following case of a quadrifaction of five letters is a matter of some difficulty.

The theory is connected with that of the invariants of a system of binary quantics.

610. On the Analytical Forms called Trees, with Application to the Theory of Chemical Combinations

[From the Report of the British Association for the Advancement of Science, (1875), pp. 257–305.]

[p. 427] I have in two papers “On the Analytical forms called Trees,” *Phil. Mag.* vol. xiii. (1857), pp. 172–176, [203], and ditto, vol. xx. (1859), pp. 374–378, [247], considered this theory; and in a paper “On the Mathematical Theory of Isomers,” ditto, vol. xlvii. (1874), p. 444, [586], pointed out its connexion with modern chemical theory. In particular, as regards the paraffins C_nH_{2n+2} , we have n atoms of carbon connected by $n - 1$ bands, under the restriction that from each carbon-atom there proceed at most 4 bands (or, in the language of the papers first referred to, we have n knots connected by $n - 1$ branches), in the form of a tree; for instance, $n = 5$, such forms (and the only such forms) are

[Three tree diagrams of carbon skeletons for $n = 5$ paraffin isomers: the linear (normal pentane), the branched (isopentane), and the centrally branched (neopentane); each carbon-atom labelled with the number of attached hydrogens.]

And if, under the foregoing restriction of only 4 bands from a carbon-atom, we connect with each carbon-atom the greatest possible number of hydrogen-atoms, as shown in the diagrams by the affixed numerals, we see that the number of hydrogen-atoms is 12 ($= 2 \cdot 5 + 2$); and we have thus the representations of three different paraffins, C_5H_{12} . It should be observed that the tree-symbol of the paraffin is

[p. 428] completely determined by means of the tree formed with the carbon-atoms, or say of the carbon-tree, and that the question of the determination of the theoretic number of the paraffins C_nH_{2n+2} is consequently that of the determination of the number of the carbon-trees of n knots, viz. the number of trees with n knots, subject to the condition that the number of branches from each knot is at most $= 4$.

In the paper of 1857, which contains no application to chemical theory, the number of branches from a knot was unlimited; and, moreover, the trees were considered as issuing each from one knot taken as a root, so that, $n = 5$, the trees regarded as distinct (instead of being as above only 3) were in all 9, viz. these were

[Nine root-tree diagrams of 5 knots each, each issuing from a designated bottom knot.]

which, regarded as issuing from the bottom knots, are in fact distinct; while, taking them as issuing each from a properly selected knot, they resolve themselves into the above-mentioned 3 forms. The problem considered was in fact that of the “general root-trees with n knots”—general, inasmuch as the number of branches from a knot was without limit; root-trees, inasmuch as the enumeration was made on the principle last referred to. It was found that for

knots	1,	2,	3,	4,	5,	6,	7,	8,
No. of trees was	1,	1,	2,	4,	9,	20,	48,	115,
	$= 1,$	$A_2,$	$A_3,$	$A_4,$	$A_5,$	$A_6,$	$A_7,$	$A_8,$

the law being given by the equation

$$(1-x)^{-1}(1-x^2)^{-A_2}(1-x^3)^{-A_3}(1-x^4)^{-A_4}\dots = 1+x+A_2x^2+A_3x^3+A_4x^4+\dots;$$

but the next following numbers A_9, A_{10}, A_{11} , the correct values of which are 286, 719, 1842, were given erroneously as 306, 775, 2009. I have since calculated two more terms, $A_{12} = 4766$, $A_{13} = 12486$.

The other questions considered in the paper of 1857 and in that of 1859 have less immediate connexion with the present paper, but for completeness I reproduce the results in a Note².

²In the paper of 1857 I also considered the problem of finding B_r , the number with r free branches, with bifurcations at least: this was given by a like formula

$$(1-x)^{-1}(1-x^2)^{-B_2}(1-x^3)^{-B_3}(1-x^4)^{-B_4}\dots = 1+x+2B_2x^2+2B_3x^3+2B_4x^4+\dots,$$

leading to $B_r = 1, 2, 6, 12, 38, 90$ for $r = 2, 3, 4, 5, 6, 7$.

In the paper of 1859, the question is to find the number of trees with a given number m of terminal knots: we have here

$$\phi_m = \frac{1}{1 \cdot 2 \cdot 3 \cdots (m-1)} \text{ coefficient of } x^{m-1} \text{ in } \Sigma,$$

giving the values $\phi_m = 1, 1, 3, 13, 75, 541, 4683, 47293, \dots$ for $m = 1, 2, 3, 4, 5, 6, 7, 8, \dots$

But if from each non-terminal knot there ascend two and only two branches, then in this case $\phi_m =$ coefficient of x^{m-1} in $\frac{1-\sqrt{1-4x}}{2}$, viz. we have the very simple form

$$\phi_m = \frac{1 \cdot 3 \cdot 5 \cdots (2m-3)}{1 \cdot 2 \cdot 3 \cdots m} \cdot 2^{m-1},$$

giving $\phi_m = 1, 1, 2, 5, 14, 42, \dots$ for $m = 1, 2, 3, 4, 5, \dots$

[p. 429] To count the trees on the principle first referred to, we require the notions of “centre” and “bicentre,” due, I believe, to Sylvester; and to establish these we require the notions of “main branch” and “altitude”: viz. in a tree, selecting any knot at pleasure as a root, the branches which issue from the root, each with all the branches that belong to it, are the main branches, and the distance of the furthest knot, measured by the number of intermediate branches, is the altitude of the main branch. Thus in the left-hand figure, taking A as the root, there are 3 main branches of the altitudes 3, 3, 1 respectively: in the right-hand figure, taking A as the root, there are 4 main branches of the altitudes 2, 2, 1, 3 respectively; and we have

[Two diagrams: left, a tree with vertex A illustrating a centre and three main branches; right, a tree with adjacent vertices A, B illustrating a bicentre.]

then the theorem that in every tree there is either one and only one centre, or else one and only one bicentre; viz. we have (as in the left-hand figure) a centre A which is such that there issue from it two or more main branches of altitudes equal to each other and superior to those of the other main branches (if any); or else (as in the right-hand figure) a bicentre AB , viz. two contiguous knots, such that issuing from A (but not counting AB), and issuing from B (but not counting BA), we have two or more main branches, one at least from A and one at least from B , of altitudes equal to each other and superior to those of the other main branches in question (if any). The theorem, once understood, is proved without difficulty: we consider two terminal knots, the distance of which, measured by the number of intermediate branches, is greater than or equal to that of any other two terminal knots; if, as in the left-hand figure, the distance is even, then the central knot A is the centre of the tree; if, as in the right-hand figure, the distance is odd, then the two central knots AB form the bicentre of the tree.

In the former case, observe that if G, H are the two terminal knots, the distance of which is $= 2\lambda$, then the distance of each from A is $= \lambda$, and there cannot be

[p. 430] any other terminal knot I , the distance of which from A is greater than λ (for, if there were, then the distance of I from G or else from H would be greater than 2λ); there cannot be any two terminal knots I, J , the distance of which is greater than 2λ ; and if there are any two knots I, J , the distance of which is $= 2\lambda$, then these belong to different main branches, the distance of each of them from A being $= \lambda$; whence, starting with I, J (instead of G, H), we obtain the same point A as centre. Similarly, in the latter case, there is a single bicentre AB .

Hence, since in any tree there is a unique centre or bicentre, the question of finding the number of distinct trees with n knots is in fact that of finding the number of centre- and bicentre-trees with n knots; or say it is the problem of the “general centre- and bicentre-trees with n knots:” general, inasmuch as the number of branches from a knot is as yet taken to be without limit; or since (as will appear) the number of the bicentre-trees can be obtained without difficulty when the problem of the root-trees is solved, the problem is that of the “general centre-trees with n knots.” It will appear that the solution depends upon and is very readily derived from that of the foregoing problem of general root-trees, so that this last has to be considered, not only for its own sake, but with a view to that of the centre-trees. And in each of the two problems we doubly divide the whole system of trees according to the number of the main branches, issuing from the root or centre as the case may be, and according to the altitude of the longest main branch or branches, or say the altitude of the tree; so that the problem really is, for a given number of knots, a given number of main branches, and a given altitude, to find the number of root-trees, or (as the case may be) centre-trees.

We next introduce the restriction that the number of branches from any knot is equal to a given number at most; viz. according as this number is 2, 3 or 4, we have, say oxygen-trees, boron-trees³, and carbon-trees respectively; and these are, as before, root-trees or centre- or

³I should have said nitrogen-trees; but it appears to me that nitrogen is of necessity 5-valent, as shown by the compound, Ammonium-Chloride, $= \text{NH}_4\text{Cl}$. Of course, the word boron is used simply to stand for a 3-valent element.

bicentre-trees, as the case may be. The case where the number is 2 presents no difficulty: in fact, if the number of knots be $= n$, then the number of root-trees is either $\frac{1}{2}(n+1)$ or $\frac{1}{2}n$; viz. $n=3$ and $n=4$, the root-trees are

[Two simple linear root-tree diagrams of 3 and 4 knots respectively.]

and the number of centre- or bicentre-trees is always 1: viz. n odd, there is one centre-tree; and n even, one bicentre-tree; it is only considered as a particular case of the general theorem. The case where the number is 3 is analytically interesting although there may not exist, for any 3-valent element, a series of hydrogen compounds

[End of pages 406–430 (PDF pp. 426–450). Paper 610 continues in subsequent pages.]